

Solving multi-step inequalities



1

- | | | | | | | | |
|---|-------------|---|-------------|---|-------------|---|--------------|
| a | $x \geq 14$ | b | $x \geq 0$ | c | $x < 5$ | d | $x > -8$ |
| e | $x > 2$ | f | $x < -13$ | g | $x \leq -3$ | h | $x \leq 0$ |
| i | $x \leq -3$ | j | $x > 12$ | k | $x \leq 16$ | l | $x \geq -24$ |
| m | $x \geq 4$ | n | $x < 19$ | o | $x < -21$ | p | $x < -14$ |
| q | $x < -2$ | r | $x \geq -1$ | s | $x \leq 7$ | t | $x > 4$ |
| u | $x \leq 45$ | v | $x > 40$ | w | $x \leq 3$ | x | $x > 27$ |

2

- | | | | | | | | |
|---|-------------------|---|--------------|---|---------|---|----------------------|
| a | $x > 2$ | b | $x \geq -12$ | c | $x > 8$ | d | $x > \frac{150}{89}$ |
| e | $x > \frac{5}{4}$ | | | | | | |

3

- | | | | | | | | |
|---|---------|----|-----|-----|-----|----|-----|
| a | $x > 0$ | | | | | | |
| b | | | | | | | |
| i | No | ii | Yes | iii | Yes | iv | Yes |

4

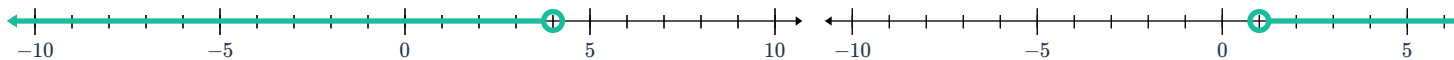
- | | | | | | | | |
|---|------------|----|-----|-----|-----|----|----|
| a | $x \leq 6$ | | | | | | |
| b | | | | | | | |
| i | Yes | ii | Yes | iii | Yes | iv | No |

5

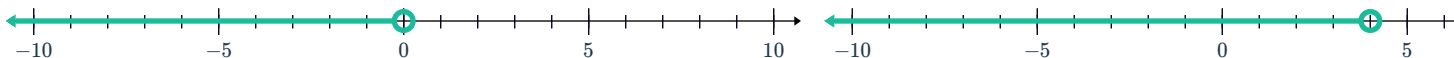
- | | | | | | | | |
|---|-------------------|----|----|-----|-----|----|-----|
| a | $x > \frac{7}{5}$ | | | | | | |
| b | | | | | | | |
| i | No | ii | No | iii | Yes | iv | Yes |

6

- | | | | | | |
|----|---------|--|--|----|---------|
| a | $x < 4$ | | | b | $x > 1$ |
| i | | | | i | |
| ii | | | | ii | |



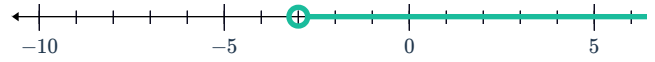
- | | | | | | |
|----|---------|--|--|----|---------|
| c | $x < 0$ | | | d | $x < 4$ |
| i | | | | i | |
| ii | | | | ii | |



e
i $x \geq -1$
ii

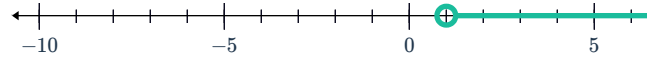
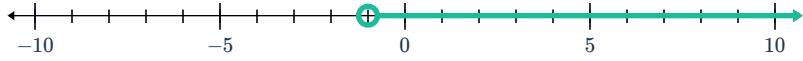


f
i $x > -3$
ii



g
i $x > -1$
ii

h
i $x > 1$
ii



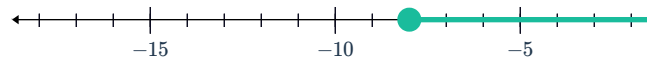
i
i $x \geq 8$
ii

j
i $x \leq 6$
ii

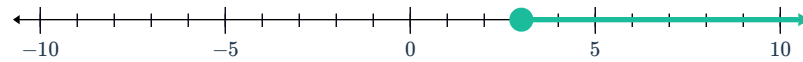


k
i $x \geq -6$
ii

l
i $x \geq -8$
ii



m
i $x \geq 3$
ii



7 Step 1 is incorrect. At this step she incorrectly combined -3 and -2 .

8 $x \geq 3$



- 9
- a $p \leq 5$
 - b $p = 5$

10 $p \leq 16.75$

11 $A \geq 65.40$

12 $x \geq 66$

13 $W \leq 8$

- 14
- a $x \leq 1630.10$
 - b \$1630.10

- 15
- a $n > \frac{40}{3}$
 - b 14 rockmelons

- 16
- a $h \geq 61.4$
 - b 62

- 17
- a $B \leq 3.4$
 - b 3

18 Lachlan, James

- 19
- a $x + 650 \geq 950$
 - b \$960
 - c Yes

- 20
- a 10 060 lb
 - b No

Additional questions

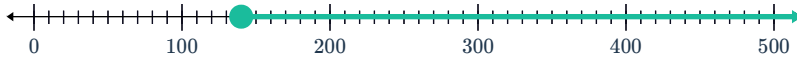
21 $3 \leq x \leq 7$ includes both end points, whereas $0 \leq x < 4$ only includes the left hand end point. $0 \leq x < 4$ includes the number 0 and every number up to 4 but not including 4. $3 \leq x \leq 7$ includes the number 3 and every number up to 7 including 7

- 22
- a x can be equal to 29 or any number greater than 29.
 - b x can be equal to any number less than -29 , not including 29.

- 23
- a $f > 2m$
 - b
 - i No
 - ii Yes
 - iii No
 - iv No

24 a $x + 410 \geq 550$

b



25 a $w \geq 8.95$

b

- i No, washing exactly 9 windows would generate \$18, which with the \$22.10 he already has is enough money to afford the equipment.
- ii No, washing 8 windows would only generate \$16 which would give him a total of \$38.10, which is not enough money to afford the equipment.
- iii No, he gets paid for washing the whole window, not parts.
- iv Yes, if Ryan washed 9 windows he would earn enough to buy the equipment. If he washed more than 9 windows, he would have more than enough money.

26 The minimum should be at -9 instead of -8 and the dot at that point should be filled as -9 is included in the solution.